ABM + RL

in the Wild

Real-World Applications of Agent-Based Modelling combined with Reinforcement Learning

From Conway to LangGraph · Week 7 Panorama · Università di Bologna

Traffic | Finance | Energy | Epidemiology | Ecology | Robotics | Social

Seven Domains

Where ABM provides the generative environment · RL provides the adaptive policy

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Traffic & Mobility

Signal control, AV fleets, SUMO+RL

📈

Financial Markets

Order execution, market-making, SFI ASM

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Energy Grids

Prosumer coordination, demand response

🦠

Epidemiology

Intervention policy, COVID modelling

🌿

Ecology & Conservation

Fishery management, predator-prey RL

🤖

Robotics & Swarms

Warehouse robots, drones, hide-and-seek

💬

Social Dynamics

Opinion formation, polarization, Ising

Common thread: emergence · phase transitions · breakdown of mean-field when heterogeneity dominates

🚗 Most Mature Domain

Traffic & Urban Mobility

Multi-agent signal coordination · autonomous vehicle routing · fleet management

Environment: SUMO Simulator

Simulation of Urban MObility — open-source microscopic traffic sim used as the ABM substrate. Each vehicle is an agent with its own OD pair, speed, and gap-acceptance policy.

RL Policy: Signal Control

Each intersection runs an independent RL agent. Reward = −(total queue length + delay). Trained with DQN / PPO. Cooperation emerges without explicit communication.

Deployed System: Hangzhou

Alibaba + DiDi deployed RL traffic signals across Hangzhou city (2019). Average travel time reduced ~12% vs. fixed-cycle baselines. Scales to 3000+ intersections.

Open Benchmarks

CityFlow, RESCO benchmark suite (2021). Standard test: 4×4 Manhattan grid, 3600 vehicles/hour. MARL outperforms actuated control by 30–40%.

VEHICLES SIMULATED (SUMO)

10,000+ agents at city scale in real time

HANGZHOU DEPLOYMENT

~12% reduction in average travel time

SIGNAL CYCLE IMPROVEMENT

30–40% over actuated baselines (RESCO)

KEY PAPERS

Wei et al. 2019 · Zheng et al. 2019 · Oroojlooy 2020

⚛ **Physics analogy:**

Traffic flow maps to driven lattice gas models. RL finds the policy that minimises a free-energy-like cost, navigating the jamming phase transition between free-flow and gridlock phases.

Financial Markets

Heterogeneous trader agents · optimal execution · market microstructure emergence

Ancestor: Santa Fe Artificial Stock Market
Holland & LeBaron (1994). Agents have evolving classifier rules; fat tails and volatility clustering emerge without hard-coding them. First ABM to reproduce stylized financial facts.

RL Application: Optimal Execution
A large order (e.g. sell 1M shares) must be split across time to avoid market impact. RL learns the optimal schedule: Almgren-Chriss is the MDP baseline. Deep RL extends to non-linear impact models.

Market Making with MARL
Multiple RL market-makers quote bid-ask prices simultaneously. Emergent spreads, inventory dynamics, and quote-stuffing behaviour arise. Used in research at JPMorgan AI Research.

Systemic Risk Modelling
EURACE agent model + RL bank capital policies (Cincotti et al.). RL identifies macro-prudential policies that reduce crisis probability — regulators take note.

ORIGINAL SFI MODEL YEAR

1994 — Holland & LeBaron et al.

STYLIZED FACTS REPRODUCED

Fat tails · volatility clustering · autocorrelation

EXECUTION SAVINGS

3–8 bps improvement over TWAP/VWAP strategies

KEY PAPERS

Almgren & Chriss 2001 · Spooner et al. 2018 · Dwarakanath 2023

⚛ Physics analogy:

Price dynamics ≈ Brownian motion with drift (Black-Scholes). But ABM reveals non-Gaussian tails — exactly as in critical systems near phase transitions. Value function ≈ free energy of the optimal trading policy.

Energy Grids & Demand Response

⚡ Strong Physics Link

Prosumer networks · distributed storage coordination · grid stability emergence

Agent Model: Prosumers

Each household/building is an agent with solar panels, battery storage, and flexible loads (EV charging, HVAC). The grid is the shared environment. Local decisions create global frequency dynamics.

RL Policy: Local Optimization

Each prosumer learns when to charge/discharge storage to minimize cost while collectively maintaining grid frequency. Reward = local bill − penalty for grid deviation. MARL with shared critic is standard.

Microgrid Coordination

CityLearn benchmark (Vazquez-Canteli & Nagy, 2019). Multi-building environment. SAC and DDPG agents trained to flatten the "duck curve" — peak demand reduction of 25–40%.

Virtual Power Plants

Aggregated MARL prosumers bid into electricity markets as a single entity. Emergent strategy: coordinated arbitrage between peak and off-peak prices without explicit communication.

CITYLEARN BENCHMARK

9 buildings · 1 year of operations · hourly decisions

PEAK DEMAND REDUCTION

25–40% via MARL prosumer coordination

CARBON REDUCTION

10–20% by shifting load to renewable generation windows

KEY PAPERS

Vazquez-Canteli 2019 · Pigott 2022 · Perez 2023

⚛ **Physics analogy:**

Energy grid ≈ coupled oscillators (Kuramoto model). Prosumer RL policies tune individual phases to maintain synchrony — stability = phase-locked state. Desynchronisation = blackout phase transition.

COVID Surge of Interest

Epidemiology & Public Health

Heterogeneous contact networks · intervention optimisation · SARS-CoV-2 policy

ABM Substrate: Individual Contact Networks

Agents = individuals with age, occupation, household, comorbidity. Contacts sampled from empirical networks (Prem et al. contact matrices). Superspreader events emerge from network topology.

RL Policy: Intervention Scheduling

Government = RL agent. Actions: lockdown level, vaccination priority, border closure. State = {I(t), H(t), ICU capacity}. Reward = −(deaths + economic cost). MDP horizon = epidemic duration.

Key Groups: Imperial College, ETH Zürich

Ferguson et al. (2020) ABM was central to UK lockdown decisions. CADENCE group (ETH) used MARL to optimise multi-country coordination. Both groups showed myopic policies outperform reactive ones.

Partial Observability Challenge

State is only partially observed (asymptomatic fraction unknown). POMDP formulation with belief states over hidden infection counts. RL outperforms rule-based policies under uncertainty.

IMPERIAL MODEL AGENTS

66M synthetic UK individuals with contact networks

POLICY IMPROVEMENT

RL schedules reduce deaths 15–30% vs. reactive lockdowns

VACCINATION STRATEGY

Priority-to-elderly RL policy robust across variant scenarios

KEY PAPERS

Ferguson et al. 2020 · Kompella et al. 2022 · Ohi 2022

⚛ Physics analogy:

SIR/SEIR ≡ mean-field approximation. ABM reveals heterogeneity effects: super-spreading, network percolation threshold. RL intervention ≡ controlling the effective reproduction number R_0 as a tunable order parameter.

Ecology & Conservation

Predator-prey dynamics · fishery collapse · wildlife corridor design

Classic Link: Predator–Prey ABM

NetLogo Wolf-Sheep model → RL predators learn hunting strategies while prey learn evasion. Population oscillations (Lotka-Volterra) emerge; RL agents push toward fixed points or chaos.

Fishery Management

Agents = fishing vessels with heterogeneous quotas and strategies. RL regulator sets total allowable catch dynamically. Key result: RL policies delay collapse vs. fixed-quota baselines (Symes 2020).

Conservation Planning

RL-optimised wildlife corridor design (Alagador & Cerdeira 2023). Agents = animal populations; environment = landscape with habitat patches. Reward = metapopulation viability index.

Invasive Species Control

Multi-agent RL for invasive plant management (Lapeyrolerie et al. 2022). RL discovers non-obvious treatment schedules: treating source patches first, not the heaviest infestations.

LOTKA-VOLTERRA CONNECTION

RL predators converge to evolutionarily stable strategy

FISHERY DELAY

RL adaptive quota delays stock collapse by 40-60%

CORRIDOR DESIGN GAIN

+25% species persistence vs. corridor optimised by hand

KEY PAPERS

Lapeyrolerie et al. 2022 · Symes 2020 · Alagador 2023

⚛ Physics analogy:

Predator-prey ≡ driven non-equilibrium system with limit cycles. SOC connection: fishery collapses are critical transitions — the RL policy keeps the system in the sub-critical regime, away from the bifurcation.

🤖 Most Visual Domain

Robotics & Swarm Systems

Warehouse robots · drone swarms · OpenAI hide-and-seek · emergent tool use

Amazon Robotics (Kiva Systems)
800+ Kiva robots coordinate in fulfilment centres. Each robot is an RL agent (path planning under collision avoidance). Throughput >2× human pick rates. The warehouse is a literal physical ABM.

OpenAI Hide-and-Seek (2019)
Multi-agent RL in physics simulator. Hiders and seekers co-evolve over 10^9 steps. Emergent phases: open play → shelter construction → box surfing → ramp sabotage. A canonical MARL curriculum result.

Drone Swarms
Decentralised MARL for UAV swarm navigation (Zhou et al. 2022). Each drone observes local neighbourhood only. Collision-free flocking, dynamic target tracking, and split-merge manoeuvres emerge.

Autonomous Vehicle Platoons
Truck platooning: RL follower vehicles minimise headway gap while maintaining string stability. MARL version extends to mixed-autonomy traffic (humans + AVs) — 40% fuel reduction in dense platoons.

AMAZON KIVA ROBOTS
750,000+ units deployed globally (2023 estimate)

OPENAI HIDE-AND-SEEK
6 emergent strategy phases over 500M episodes

PLATOON FUEL REDUCTION
~40% at 8–10 m headway in dense mixed-AV traffic

KEY PAPERS
Baker et al. 2019 · Zhou et al. 2022 · Jang et al. 2019

▶ VIDEO PLACEHOLDER — insert clip here (swarm / hide-and-seek / warehouse robots)

Social Dynamics & Opinion Formation

Direct Ising Link

Network RL agents · polarisation · influence maximisation · Ising connections

Voter Model → RL Voters

Classic voter model on graphs: agents copy neighbours. RL extension: agents choose which neighbour to copy, or when to switch. Polarisation emerges when RL agents weight in-group ties more heavily.

Influence Maximisation

RL agent = social media platform deciding which content to boost. Reward = engagement / reach. Emergent effect: recommendation RL agents inadvertently create echo chambers — demonstrated on Reddit & Twitter data.

Disinformation Dynamics

Contra-agent ABM: RL fact-checker selects which viral posts to correct. Reward = −(misinformation reach). RL finds non-obvious strategies: ignoring micro-influencers, targeting bridge nodes.

Pandemic Opinion + Behaviour

Opinion ABM coupled to epidemiological ABM. RL government optimises messaging strategy alongside intervention. Opinion dynamics modulate compliance → feed back into R_0. Non-linear coupling.

POLARISATION ONSET

Sharp transition at rewiring probability $p \approx 0.04$ (SBM graphs)

ECHO CHAMBER FORMATION

Emerges within 10^3 RL recommendation steps on scale-free nets

FACT-CHECKING GAIN

RL reduces misinformation reach 35% vs. random correction

KEY PAPERS

Pham et al. 2022 · Sirbu 2019 · Törnberg 2022 · Lu 2023

⚛ Physics analogy:

Opinion formation ≡ Ising model on a graph. Ferromagnetic phase = consensus; paramagnetic = fragmented discourse. RL recommendation policies effectively tune the coupling J — and can drive the system through the phase transition.